

November 20th Target Arm Survey

Scan 1 & 2

Purpose was to see how the pitch and the yaw of foil surfaces varied from foil to foil and also foil to ladder. Scan 1 was for the ladder in the fully retracted position (on switch) and scan 2 was in the fully extended position (on switch). Pitch is the angle we do not control while yaw is.

Scan 1	Ladder	Foil 1	Foil 4
Yaw	3.6	3.58	-4
Pitch	90.1	90.6	90.8

Scan 2	Ladder	Foil 1	Foil 3	Foil 4
Yaw	3.15	3.1	2.1	2.4
Pitch	90.2	90.8	90.1	91.4

Results with different sampling areas to reconstruct surface (used ~5 points)

Scan 2	Foil 4	Foil 4	Foil 4	Foil 4
Yaw	2.4	3.4	2.67	3.3
Pitch	91.4	90.9	90.4	90.7

Conclusion

- The pitch of the foils will vary up to 1 degree despite ladder being of .1 -.2 degrees.
- The yaw of the foil surface will vary up to one degree with respect to the ladder.
- We will need the foil surface pitch and yaw with respect to the ladder for each foils 1-4.
- The last scan of the day was done to get this data.

Scans 3 – 8

Purpose- Find linear encoder values for foil centers and parked position

Note-This is where we start using the software for the first time. We told the software to move in degrees and mm. Encoder values were displayed on the screen.

Procedure

- retracted arm until reached negative limit
- move about 2mm from limit. This is parked position.
- scan 3 told us foil center 1 was 43.47 mm from beam line
- told software to move 43.47 mm
- scan 4 told us we were .3 mm below the beam line
- we tried to move .03 mm to center of foil but scan 5 said we were .07mm off. I am assuming we moved the wrong way since we were still getting acquainted with the software
- we then continued to move in increments of 37mm to survey the other foil centers

	Foil Center below beamline (mm)	Rotational Encoder (Volts)
On - limit		3.055
Parked		3.025
Foil 1	.03	2.364
Foil 2	.25	1.809
Foil 3	.07	1.250
Foil 4	.09	0.679

Conclusion- This gives the linear encoder values to use for the four foil positions and parked position. We can either use these values or adjust them to be closer to the foil center.

	Linear Encoder (Volts)
Parked	3.025
Foil 1 Center	2.364
Foil 2 Center	1.813
Foil 3 Center	1.251
Foil 4 Center	.680

Scans 9-15

Purpose- To determine rotational encoder values

Procedure

- rotate to negative limit
- forward till off switch (parked position)
- survey told us we were -7.9 degrees from beam axis
- rotated 4 degrees
- survey told us we were 3.9 degrees from beam axis
- rotate 3.9 degree
- survey told us -.2 degrees from beam axis
- continued to rotate 4 or 5 degrees surveying each time
- scan 15 we tried to get to 0 degrees (ended at .2) using predicted encoder value

Data all taken with ladder in foil 1 on beam position

Degrees	-7.9	-3.9	-.2	.2	3.9	8	13.1
RENC	4.150	4.353	4.558	4.565	4.750	4.960	5.206

Scan 15 -18

Purpose- To see how the yaw angle of the ladder changes as the ladder moves into different target positions

Procedure

-With the ladder at an encoder reading of 4.565 we went from Foil 1 to Foil 4 to Foil 1 to Foil 4

Results

	Foil 1	Foil 4	Foil 1	Foil4
Yaw Angle of Ladder	.2	-.6	.2	-.6

Conclusion

The yaw angle of the ladder changes .8 degrees from foil 1 position to foil 4 position

Scan 19-21

Purpose-Scans 9-15 gave us rotational encoder values at beam on Foil 1 position. We did the same thing for the ladder at beam on Foil 4 position

Yaw Angle of Ladder	-8.1	-.1	13.1
RENC	4.185	4.600	5.262

Scan 22-23

Purpose-See how yaw angle of ladder changes with ladder position

Results-All data taken with RENC = 4.566

	Foil 1	Foil 2	Foil 3	Foil4
Yaw Angle Of Ladder	.2	0	-.4	-.6

Conclusion-

The rotational encoder value for target perpendicular to beam will be different for each target position. This is without considering the foil surface angle with respect to the ladder

Scan 24-26

Purpose- To see how physically bumping the target effects the yaw

Scan 24 was taken after Bill bumped the ladder. The angle changed .1 degrees.

Scan 25 was taken after Ethan bumped the ladder. The angle went from -.1 to -.7.

Scan 26 was taken after rotating ladder to parked position and then back to encoder value the ladder was at before the bump. The angle was still -.7 degrees.

Conclusion

-Bumping the ladder leads to a change in the calibration of up to .7 degrees that can not be restored by rotating the ladder

Scan 27

The last scan of the day was a scan of the ladder and each foil surface. This will give us each foil surface's yaw angle with respect to the ladder. This data still has to be acquired from the survey team.